# AMEV

# ADAPTIVE MECHANICAL ENERGY VEHICLE

PROJECT CONCEPT & TECHNICAL OVERVIEW

| | |
|---|---|
| **Project type** | Research-oriented vehicle concept |
| **Development stage** | Concept / prototype development |
| **Core areas** | Pneumatics · Embedded sensing · Mechanical systems |
| **Document purpose** | Public project documentation for a GitHub repository |

**PROJECT STATUS**
**AMEV is a concept-stage research project. Performance, energy-recovery, efficiency, range, environmental-benefit, and novelty claims require appropriate testing and validation.**

GitHub-ready documentation • Prepared for technical review and iterative development

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## 01 | Executive Summary

AMEV (Adaptive Mechanical Energy Vehicle) is a research-oriented vehicle concept that investigates compressed-air propulsion supported by adaptive pressure control, sensor-based monitoring, and mechanical energy-utilisation studies. The proposed system stores compressed air in a suitably rated vessel, regulates the air supply, and directs it to a pneumatic motor or actuator connected to a mechanical drivetrain.

An ESP32-based control unit is intended to monitor pressure, temperature, and rotational speed. The concept also investigates a flywheel-based approach to kinetic-energy recovery during deceleration and experimental PRC/APERS modules for studying residual pneumatic exhaust-air utilisation. These remain research concepts, not validated performance improvements.

AMEV will be evaluated through measured energy input and output, pressure, airflow, temperature, speed, and distance. Its initial goal is a safe proof-of-concept platform; any future mobility application depends on performance data, safety review, cost analysis, and comparison with alternatives.

## 02 | Problem Statement

Conventional road vehicles rely on combustion fuels or battery-electric systems, each with requirements related to energy supply, infrastructure, cost, maintenance, and environmental impact. Energy losses can occur through drivetrain friction, braking, air expansion, leakage, and other inefficiencies.

AMEV explores whether a controlled compressed-air propulsion platform, combined with monitoring and experimental recovery concepts, can provide useful insights into alternative vehicle design and energy management. It is not presented as a proven replacement for petrol, diesel, or battery-electric vehicles.

## 03 | Proposed Solution

- **Compressed-air supply:** Certified pressure-rated vessel or controlled direct-compressor test arrangement with pressure regulation and over-pressure protection.
- **Pneumatic propulsion:** Pneumatic motor or actuator converts pressure energy into mechanical motion.
- **Mechanical drivetrain:** Couplings, shafts, gears, chain, or belt transmit output to the drive axle.
- **Adaptive monitoring:** ESP32 connected to pressure, temperature, and RPM/speed sensors for data acquisition and appropriate control functions.
- **Flywheel investigation:** Guarded subsystem to study kinetic-energy storage and recovery during deceleration.
- **PRC/APERS exploration:** Experimental module for studying possible residual pneumatic exhaust-air utilisation; effectiveness and novelty remain to be demonstrated.

## 04 | Working Principle

| Stage | Function |
|---|---|
| 1. Air supply | Compressed air is supplied from a pressure-rated source; component ratings and pressure limits must be respected. |
| 2. Pressure regulation | A regulator, gauge, and appropriately rated relief/safety devices help maintain controlled operating pressure. |
| 3. Flow control | A control valve meters air to the pneumatic motor or actuator. |
| 4. Mechanical drive | Motor/actuator output passes through a coupling and drivetrain to the axle and wheels. |
| 5. Monitoring | Sensors and ESP32 record pressure, temperature, RPM/speed, and selected test parameters. |
| 6. Experimental recovery | Flywheel and PRC/APERS concepts are tested separately under controlled conditions and compared with baseline operation. |

## 05 | Objectives & Differentiating Elements

- Combine compressed-air propulsion research with ESP32-based sensor monitoring in a modular platform.
- Use measured data to study pressure, temperature, airflow, RPM, speed, and energy losses rather than relying on theoretical claims alone.
- Investigate flywheel-based kinetic-energy recovery as a separate experimental subsystem.
- Evaluate PRC/APERS as an exploratory pneumatic exhaust-air utilisation concept, subject to testing and prior-art review.
- Keep propulsion, monitoring, and recovery subsystems modular so they can be tested independently.

The individual technologies have established prior art. The proposed integration and any claimed novelty require a systematic literature and patent search before intellectual-property claims are made.

## 06 | Technical Architecture

| Subsystem | Proposed components | Purpose |
|---|---|---|
| Pneumatic supply | Pressure-rated source/tank, regulator, gauge, relief valve, rated tubing and fittings | Controlled air supply with safety protection |
| Propulsion | Pneumatic motor or actuator, flow-control valve, exhaust path | Convert pneumatic energy into mechanical motion |
| Drivetrain | Coupling, shaft/gearing/chain/belt, axle and wheels | Transmit mechanical output |

| Subsystem | Proposed components | Purpose |
| --- | --- | --- |
| Monitoring/control | ESP32, pressure and temperature sensors, RPM/speed sensor, wiring and data logging | Record conditions and support controlled testing |
| Energy-recovery research | Guarded flywheel assembly; experimental PRC/APERS module | Evaluate recovery and residual-air utilisation |

# 07 | Implementation Methodology

- **Phase 1 — Requirements and safety review:** Define the test platform, load, speed targets, operating limits, and safe test conditions.
- **Phase 2 — Design and calculations:** Estimate torque, wheel speed, airflow, pressure range, drivetrain ratio, and energy budget.
- **Phase 3 — Component selection:** Choose compatible pneumatic, drivetrain, and sensor components using verified specifications.
- **Phase 4 — Bench prototype:** Assemble and test the pneumatic drive without a person riding on the vehicle.
- **Phase 5 — Monitoring integration:** Connect ESP32 sensors, log measurements, and implement suitable shutdown logic.
- **Phase 6 — Controlled testing:** Measure pressure, airflow, temperature, RPM, speed, distance, compressor/refill energy, and mechanical output.
- **Phase 7 — Recovery experiments:** Evaluate flywheel and PRC/APERS subsystems independently against a baseline.
- **Phase 8 — Review and refinement:** Analyse repeatable results, document limitations, and update the design.

# 08 | Feasibility, Risks & Mitigation

| Risk / limitation | Mitigation / validation approach |
|---|---|
| Low energy density and limited compressed-air runtime | Calculate energy demand; measure airflow, pressure decay, and runtime; compare alternatives. |
| Pressure vessel or fitting failure | Use certified, rated components, relief protection, and qualified review. Never use ordinary bottles as pressure vessels. |
| Air leakage and pressure drop | Use compatible rated fittings/tubing and conduct leak checks before operation. |
| Cooling during expansion | Monitor temperature and pressure during repeated controlled tests. |
| Flywheel hazards from rotating mass | Use suitable balancing, robust guarding, secure mounts, and restricted test conditions. |
| Unverified PRC/APERS benefit | Treat as experimental; compare input/output measurements and report negative or inconclusive results. |
| Environmental impact depends on compression energy | Record energy used to compress/refill air and consider its source in environmental assessment. |

# 09 | Testing & Validation Plan

The first prototype should be validated on a secured bench or in a controlled test area. The test log should include date, configuration, source pressure, airflow, air temperature, motor RPM, wheel speed, distance or duration, load, leakage observations, and compressor/refill energy. Repeat tests under comparable conditions and distinguish measured values from estimates.

- **Baseline test:** Pneumatic propulsion and drivetrain only.
- **Monitoring test:** Verify sensor readings against suitable reference instruments.
- **Flywheel test:** Compare deceleration behaviour and energy balance with and without the flywheel subsystem.
- **PRC/APERS test:** Quantify exhaust-air conditions and any useful output; do not assume recovery is beneficial until measured.

- **Reporting:** Include uncertainty, losses, limitations, failed tests, and inconclusive findings—not only best-case results.

## 10 | Expected Outcomes

- Documented proof-of-concept for studying compressed-air propulsion and mechanical transmission.
- ESP32-based dataset showing how pressure, temperature, RPM, and speed vary during operation.
- Experimental evidence to assess whether flywheel recovery provides measurable benefit in the selected setup.
- A test framework for evaluating PRC/APERS without assuming a positive outcome.
- Practical learning in IoT sensing, mechanical design, control, safety, and sustainable mobility research.

Environmental benefit is not guaranteed by the absence of onboard combustion alone. Compression energy, total efficiency, materials, and maintenance must be considered in a full assessment.

## 11 | Development Roadmap

| Period | Milestone | Deliverable |
|---|---|---|
| Month 1 | Requirements, safety review, calculations, and prior-art search | System specification and initial design |
| Months 2–3 | Procure compatible components and assemble bench-level propulsion | Bench prototype and basic movement test |
| Month 4 | Integrate ESP32 sensors and data logging | Monitoring logs and sensor validation |
| Month 5 | Conduct repeatable baseline and subsystem tests | Test dataset and energy-balance analysis |
| Month 6 | Evaluate flywheel and PRC/APERS experiments; refine design | Final test report, limitations, and next-stage plan |

## 12 | Preliminary Prototype Budget

Indicative planning estimate from the concept document. Prices must be confirmed with supplier quotations and final component specifications.

| Category | Component | Qty / basis | Estimated cost |
|---|---|---|---|
| Pneumatic propulsion | Pressure-rated compressed-air tanks | 2 units | ■36,000 |
| Pneumatic propulsion | Pneumatic air motor | 1 unit | ■45,000 |
| Pneumatic propulsion | Pressure regulator and control valves | 1 set | ■15,000 |
| Pneumatic propulsion | Pressure-rated tubing and fittings | 1 set | ■8,000 |
| Pneumatic propulsion | Pressure sensors | 2 units | ■5,000 |
| Energy recovery | Flywheel assembly | 1 assembly | ■25,000 |
| Energy recovery | Flywheel housing / guard | 1 set | ■8,000 |
| Energy recovery | Optional generator / test load | 1 unit | ■6,000 |
| Energy recovery | Coupling and bearings | 1 set | ■5,000 |
| Electronics and control | ESP32 development board | 1 unit | ■1,500 |
| Electronics and control | Temperature, RPM, and other sensors | 1 set | ■5,000 |

| Category | Component | Qty / basis | Estimated cost |
|---|---|---|---|
| Electronics and control | Motor / valve controller interface | 1 set | ■4,000 |
| Electronics and control | Display and data logging | 1 set | ■2,500 |
| Electronics and control | Wiring, connectors, and PCB | 1 lot | ■5,000 |
| Chassis and fabrication | Chassis frame fabrication | 1 frame | ■40,000 |
| Chassis and fabrication | Wheels and tyres | 4 units | ■20,000 |
| Chassis and fabrication | Steering and suspension | 1 set | ■15,000 |
| Chassis and fabrication | Seat and body panels | 1 set | ■10,000 |
| Testing and safety | Tools and consumables | 1 lot | ■10,000 |
| Testing and safety | Safety equipment / guarding | 1 lot | ■7,500 |
| Testing and safety | Prototype testing and iteration | Allowance | ■25,000 |
| | **TOTAL ESTIMATED BUDGET** | | **■4,72,500** |

**Budget exclusions and uncertainty:** Taxes, shipping, major redesign, institutional facilities, specialist certification, custom fabrication, and unforeseen costs may change the final amount. This is an estimate, not a supplier quotation. PRC/APERS development may require additional funding.

# 13 | Limitations & Open Questions

- The concept has not demonstrated measured efficiency, range, or environmental superiority.
- Compressed-air energy density and expansion-related cooling may limit performance.
- Flywheel recovery benefit must be established through measured energy balance.
- PRC/APERS effectiveness and novelty have not been demonstrated.
- The integrated design and any claimed novelty require systematic literature and patent review.
- Component ratings, calculations, risk assessment, and experimental results must be updated as development proceeds.

# 14 | References & Technical Resources


- Espressif Systems — ESP32 documentation and technical resources.
- Saidur et al., “A review on compressed-air energy use and energy savings,” *Renewable and Sustainable Energy Reviews* (2010). DOI: 10.1016/j.rser.2009.07.012.
- NITI Aayog — official website.
- Manufacturer datasheets and safety documentation for the selected pressure vessel, regulator, relief valve, pneumatic motor, sensors, and drivetrain components should be added once exact components are selected.


**Repository note: This document describes a concept-stage research project, not a validated vehicle design or a guarantee of performance. As development progresses, add verified component specifications, calculations, circuit/mechanical diagrams, test data, photographs, and measured results.**